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MZsearch: A Python-Based Compound Identification Tool for GC-MS and LC-MS-Based Metabolomics

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Summary

A primary goal in metabolomics, the qualitative and quantitative study of metabolites, is to identify chemical compounds in a given sample. This often involves identifying and quantifying metabolic biomarkers for use in diagnosing, treating, and stratifying the risk of various diseases. A central tool for compound identification in metabolomics is mass spectrometry, which produces the mass spectrum for a given chemical compound. Each peak in the mass spectrum represents an ion fragment, with one axis representing mass to charge ratio (m/z) and the other axis representing relative intensity. A popular method used to identify an unknown chemical compound based on its mass spectrum is spectral library matching. This process involves computing a measure of similarity between the mass spectrum of the unknown chemical compound and each mass spectrum in a reference library. The unknown chemical compound is then identified as the one from the reference library whose mass spectrum is most similar to that of the unknown compound. We present MZsearch, a command-line tool written in Python that implements spectral library matching. The MZsearch tool offers a range of spectrum preprocessing transformations and similarity measures, and it is capable of analyzing data generated from gas chromatography-mass spectrometry (GC-MS) and liquid chromatography-mass spectrometry (LC-MS).

Statement of need

There exist several open-source software tools for working with mass spectrometry data. The Python package matchms allows one to perform spectral library matching for both GC-MS and LC-MS and provides an application programming interface (API) for users to define their own spectrum preprocessing transformations and similarity measures (Huber et al., 2020). The pyOpenMS tool, which is a Python wrapper of the OpenMS tool, is for analyzing LC-MS data and provides common spectrum preprocessing transformations such as filtering based on m/z and intensity values (Röst et al., 2014, 2016). The Python package spectrum_utils provides functionality for preprocessing and visualizing mass spectrometry data (Bittremieux, 2020). The R package metID is a tool for LC-MS-based compound identification with a focus on allowing users to combine their in-house database(s) with public databases (Shen et al., 2022). The R Shiny package ShinyMetID provides users six similarity measures (Cosine, Weighted Cosine, Stein & Scott, Discrete Fourier Transformation, Discrete Wavelet Transformation, and Semi-Partial Correlation) and a graphical user interface (GUI) to perform spectral library matching to identify chemical compounds from GC-MS data (Oh et al., 2023).

In a GC-MS experiment, the resulting mass spectra typically have m/z values as integers. In contrast, LC-MS experiments produce mass spectra with high resolution m/z values, often with many decimal places. This necessitates several spectrum preprocessing steps unique to

LC-MS data analysis, namely centroiding (i.e., merging peaks that are 'close' with respect to their m/z values) and matching (i.e., aligning the m/z values so that the query spectrum and reference spectrum have the same length). In addition to these canonical spectrum preprocessing transformations for LC-MS data, weight factor transformations and low-entropy transformations have been proposed to improve the performance of compound identification for both GC-MS and LC-MS data (Dlugas et al., 2024a; Kim et al., 2012; Li et al., 2021). The Shannon Entropy Similarity Measure has been shown to outperform the Cosine Similarity Measure with respect to LC-MS data (Li et al., 2021). A generalization of the Shannon Entropy Similarity Measure, the Tsallis Entropy Similarity Measure, slightly outperforms the Shannon Entropy Similarity Measure in both GC-MS and LC-MS data (Dlugas et al., 2024a). This recent study has further demonstrated that the order of preprocessing transformations is also critical to achieving higher accuracy in compound identification. However, to our knowledge, there is no tool that includes the recently-developed, high-performance entropy-based similarity measures or considers the order of the spectrum preprocessing transformations, which are important for effective metabolomics studies.

To address the lack of spectral library matching software that considers both the order of spectrum preprocessing steps and novel entropy-based similarity measures, MZsearch was developed. The developed MZsearch is a command-line tool for performing spectral library matching on either GC-MS or LC-MS data. It allows users to construct their own spectrum preprocessing order using spectrum preprocessing transformations such as weight factor and low-entropy transformations. Additionally, the MZsearch tool includes a novel entropy-based similarity measure, the Rényi Entropy Similarity Measure, enabling users to choose among four similarity measures: the commonly-used Cosine Similarity Measure (Stein & Scott, 1994), the Shannon Entropy Similarity Measure (Li et al., 2021), the recently-developed Tsallis Entropy Similarity Measure (Dlugas et al., 2024a), and the novel Rényi Entropy Similarity Measure.

Functionality

Spectrum Preprocessing Transformations

Functionality implementing the following spectrum preprocessing transformations is offered in MZsearch:

- Filtering: Given user-defined parameters (mz_min,mz_max), (int_min,int_max) and spectrum I with m/z values $(m_1, m_2, ..., m_n)$ and intensities $(x_1, x_2, ..., x_n)$, the transformed spectrum $I^\star$ consists of the peaks (m_i, x_i) in I such that mz_min $\leq m_i \leq$ mz_max and int_min $\leq x_i \leq$ int_max.
- Weight Factor Transformation: Given a pair of user-defined weight factor parameters (a,b) and spectrum I with m/z values $(m_1, m_2, ..., m_n)$ and intensities $(x_1, x_2, ..., x_n)$, the transformed spectrum $I^\star$ has the same m/z values as I and has intensities given by $I^\star := (m_1^{\mathsf{a}} \cdot x_1^{\mathsf{b}}, m_2^{\mathsf{a}} \cdot x_2^{\mathsf{b}}, ..., m_n^{\mathsf{a}} \cdot x_n^{\mathsf{b}})$.
- Low-Entropy Transformation: Given a user-defined low-entropy threshold parameter T and spectrum I with intensities $(x_1, x_2, ..., x_n)$, $\sum_{i=1}^{n} x_i = 1$, and Shannon entropy $H_{Shannon}(I) = -\sum_{i=1}^{n} x_i \cdot ln(x_i)$, the transformed spectrum intensities $I^\star = (x_1^\star, x_2^\star, ..., x_n^\star)$ are such that, for all $i \in \{1, 2, ..., n\}$, $x_i^\star = x_i$ if $H_{Shannon}(I) \geq T$ and $x_i^\star = x_i^{\frac{1+H_{Shannon}(I)}{1+T}}$ if $H_{Shannon}(I) < T$.
- Centroiding (only applicable to LC-MS data): Given a user-defined window-size parameter $w_{centroiding}$ and a spectrum I with m/z values $(m_1, m_2, ..., m_n)$ and intensities $(x_1, x_2, ..., x_n)$, the transformed spectrum $I^\star$ merges adjacent peaks $(m_i, x_i), (m_{i+1}, x_{i+1})$ into the peak $(\frac{m_i \cdot x_i + m_{i+1} \cdot x_{i+1}}{x_i + x_{i+1}}, x_i + x_{i+1})$ if $|m_i - m_{i+1}| < w_{centroiding}$ for $i \in \{1, 2, ..., n-1\}$. This centroiding procedure

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generalizes to more than two peaks whose m/z values are within distance $w_{centroiding}$ of each other.

- Noise Removal: Given a user-defined noise removal parameter r and a spectrum I with intensities $(x_1, x_2, ..., x_n)$, noise removal removes peaks from I with $x_j < r \cdot \max(\{x_1, x_2, ..., x_n\})$ for $j \in \{1, 2, ..., n\}$.

- Matching (only applicable to LC-MS data): Given a user-defined window-size parameter $w_{matching}$ and two spectra I, J with m/z ratios $(a_1, a_2, ..., a_n), (b_1, b_2, ..., b_m)$ and intensities $(x_1, x_2, ..., x_n), (y_1, y_2, ..., y_m)$, respectively, of which we would like to measure the similarity between, the matching procedure outputs two spectra $I^\star, J^\star$ containing the same number of peaks with $I^\star$ and $J^\star$ having transformed intensities and identical m/z ratios. Specifically, for a given peak (a_i, x_i) of I, if there are no peaks (b_j, y_j) in J with $|a_i - b_j| < w_{matching}$, then the peak (a_i, x_i) remains in $I^\star$ and the peak $(a_i, 0)$ is included in $J^\star$. If there is at least one peak (b_j, y_j) with $|a_i - b_j| < w_{matching}$, then the peak (a_i, x_i) remains in $I^\star$ and the peak $(a_i, \sum_{j \text{ such that } |a_i - b_j| < w_{matching}} b_j)$ is included in $J^\star$. This procedure is applied when transposing the roles of I and J as well.

- Normalization: Prior to computing entropy - regardless of whether in the context of performing the low-entropy transformation or computing an entropy-based similarity score - the intensities of each spectrum must be normalized to sum to 1 in order to represent a probability distribution. To normalize a given spectrum I with intensities $(x_1, x_2, ..., x_n)$ into spectrum $I^\star$ with intensities $(x_1^\star, x_2^\star, ..., x_n^\star)$ such that $\sum_{i=1}^n x_i^\star = 1$, two methods are offered:

  - Standard: $x_i^\star = \frac{x_i}{\sum_{i=1}^n x_i}$.

  - Softmax: $x_i^\star = \frac{e^{x_i}}{\sum_{i=1}^n e^{x_i}}$ where $e \approx 2.72$ is Euler's constant.

The flowchart in Figure 1 depicts the overall workflow of MZsearch.

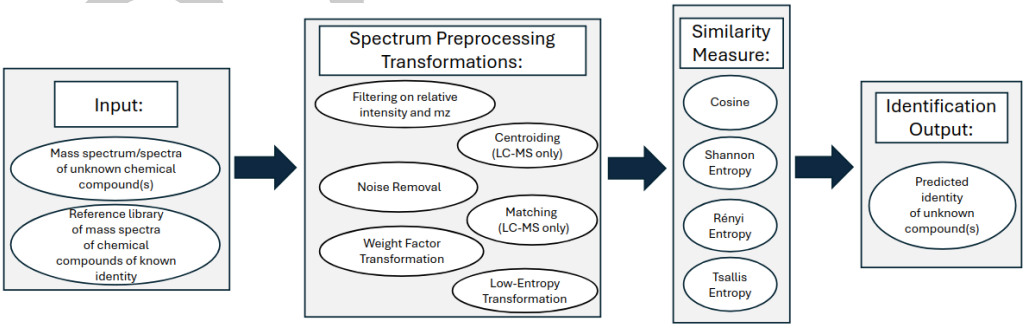

Figure 1: Workflow of MZsearch.

Similarity Measures

Given a pair of processed spectra intensities $I = (a_1, a_2, ..., a_n), J = (b_1, b_2, ..., b_n) \in \mathbb{R}^n$ with $0 \leq a_i, b_i \leq 1$ for all $i \in \{1, 2, ..., n\}$ and $\sum_{i=1}^n a_i = \sum_{i=1}^n b_i = 1$, MZsearch provides functionality for computing the following similarity measures:

- Cosine Similarity Measure:

$$S_{Cosine}(I, J) = \frac{I \circ J}{|I|_2 \cdot |J|_2}$$

  where multiplication in the numerator refers to the dot product $I \circ J = a_1 b_1 + a_2 b_2 + ... + a_n b_n$ of I and J and multiplication in the denominator refers to multiplication of the L^2-norm of I and J, $|I|_2 = \sqrt{a_1^2 + a_2^2 + ... + a_n^2}, |J|_2 = \sqrt{b_1^2 + b_2^2 + ... + b_n^2}$.

- Shannon Entropy Similarity Measure:

$$S_{Shannon}(I,J) = 1 - \frac{2 \cdot H_{Shannon}\left(\frac{I+J}{2}\right) - H_{Shannon}(I) - H_{Shannon}(J)}{ln(4)},$$
$$H_{Shannon}(I) = -\sum_{i=1}^{n} a_i \cdot ln(a_i)$$

- Tsallis Entropy Similarity Measure:

$$S_{Tsallis}(I,J,q) = 1 - \frac{2 \times H_{Tsallis}(I/2 + J/2, q) - H_{Tsallis}(I,q) - H_{Tsallis}(J,q)}{N_{Tsallis}(I,J,q)},$$
$$N_{Tsallis}(I,J,q) := \frac{\sum_{i=1}^{n}\left(2\left(\frac{a_i}{2}\right)^q + 2\left(\frac{b_i}{2}\right)^q - a_i^q - b_i^q\right)}{1-q},$$
$$H_{Tsallis}(I,q) = \frac{\left(\sum_{i=1}^{n} a_i^q\right) - 1}{1-q},$$
$$q \neq 1,\ q > 0$$

- Rényi Entropy Similarity Measure:

$$S_{Renyi}(I,J,q) = 1 - \frac{2 \times H_{Renyi}(I/2 + J/2, q) - H_{Renyi}(I,q) - H_{Renyi}(J,q)}{N_{Renyi}(I,J,q)},$$
$$N_{Renyi}(I,J,q) := \left(\frac{1}{1-q}\right)\left(2 \times ln\left(\sum_i (a_i/2)^q + \sum_j (b_j/2)^q\right) - ln(\sum_i a_i^q) - ln(\sum_i b_i^q)\right)$$
$$H_{Renyi}(I,q) = \frac{1}{1-q} ln(\sum_{i=1}^{n} a_i^q),$$
$$q \neq 1,\ q > 0$$

Usage

MZsearch has three main capabilities: (i) converting the raw data to the necessary format for spectral library matching, (ii) running spectral library matching to identify compounds based on their mass spectrometry data, and (iii) plotting a query spectrum vs. a reference spectrum before and after preprocessing transformations.

These tasks are implemented separately for (i) GC-MS and (ii) LC-MS data due to the different spectrum preprocessing transformations. To see all parameters for any of the five main scripts (build_library.py, spec_lib_matching_lcms.py, spec_lib_matching_gcms.py, plot_spectra_lcms.py, plot_spectra_gcms.py), run:

```
python build_library.py -h
python spec_lib_matching_lcms.py -h
python spec_lib_matching_lcms.py -h
python plot_spectra_lcms.py -h
python plot_spectra_gcms.py -h
```

The user can use their own libraries, and by default, MZsearch provides GC-MS and LC-MS reference libraries that have been trimmed. The recommended GC-MS and LC-MS reference libraries are available at the Zenodo database (Dlugas et al., 2024b). The GC-MS library was extracted from the National Institute of Standards and Technology WebBook, while the LC-MS library was obtained from Global Natural Products Social Molecular Networking.

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Get query/reference library from mgf, mzML, or cdf file

To obtain a query/reference library in the necessary format for spectral library matching from an mgf, mzML, or cdf file, one can use:

```
python build_library.py \
  --input_path path_to_input_mgf_or_mzML_or_cdf_file \
  --output_path path_to_output_csv_file
```

Run spectral library matching

To run spectral library matching on LC-MS/GC-MS data, one can use:

```
python spec_lib_matching_lcms.py \
  --query_data path_to_query_lcms_CSV_file \
  --reference_data path_to_reference_lcms_CSV_file \
  --similarity_measure cosine \
  --spectrum_preprocessing_order FCNMWL \
  --high_quality_reference_library False \
  --mz_min 0\
  --mz_max 999999999999\
  --int_min 0\
  --int_max 999999999999\
  --window_size_centroiding 0.5 \
  --window_size_matching 0.5 \
  --noise_threshold 0 \
  --wf_mz 0 \
  --wf_intensity 1 \
  --LET_threshold 0 \
  --entropy_dimension 1.1 \
  --normalization_method standard \
  --n_top_matches_to_save 1 \
  --print_id_results False \
  --output_identification path_to_lcms_identification_results_CSV \
  --output_similarity_scores path_to_CSV_of_all_lcms_similarity_scores

python spec_lib_matching_gcms.py \
  --query_data path_to_query_gcms_CSV_file \
  --reference_data path_to_reference_gcms_CSV_file \
  --similarity_measure cosine \
  --spectrum_preprocessing_order FNLW \
  --high_quality_reference_library False \
  --mz_min 0\
  --mz_max 999999999999\
  --int_min 0\
  --int_max 999999999999\
  --wf_mz 0 \
  --wf_intensity 1 \
  --entropy_dimension 1.1 \
  --normalization_method standard \
  --n_top_matches_to_save 1 \
  --print_id_results False \
  --output_identification path_to_gcms_identification_results_CSV \
  --output_similarity_scores path_to_CSV_of_all_gcms_similarity_scores
```

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Plot a query spectrum against a reference spectrum before and after spectrum preprocessing transformations

To plot a query spectrum vs. a reference spectrum before and after preprocessing transformations, run:

```
python plot_spectra_lcms.py \
  --query_data path_to_query_lcms_CSV_file \
  --reference_data path_to_reference_lcms_CSV_file \
  --likely_reference_IDs path_to_likely_reference_IDs_CSV_file \
  --query_spectrum_ID insert_single_ID_from_first_column_of_query_data \
  --reference_spectrum_ID insert_single_ID_from_first_column_of_reference_data \
  --similarity_measure cosine \
  --spectrum_preprocessing_order FCNMWL \
  --high_quality_reference_library False \
  --mz_min 0\
  --mz_max 999999999999\
  --int_min 0\
  --int_max 999999999999\
  --window_size_centroiding 0.5 \
  --window_size_matching 0.5 \
  --noise_threshold 0 \
  --wf_mz 0 \
  --wf_intensity 1 \
  --LET_threshold 0 \
  --entropy_dimension 1.1 \
  --normalization_method standard \
  --save_plots path_to_output_PDF_file

python plot_spectra_gcms.py \
  --query_data path_to_query_gcms_CSV_file \
  --reference_data path_to_reference_gcms_CSV_file \
  --likely_reference_IDs path_to_likely_reference_IDs_CSV_file \
  --query_spectrum_ID insert_single_ID_from_first_column_of_query_data \
  --reference_spectrum_ID insert_single_ID_from_first_column_of_reference_data \
  --similarity_measure cosine \
  --spectrum_preprocessing_order FNLW \
  --high_quality_reference_library False \
  --mz_min 0\
  --mz_max 999999999999\
  --int_min 0\
  --int_max 999999999999\
  --wf_mz 0 \
  --wf_intensity 1 \
  --LET_threshold 0 \
  --entropy_dimension 1.1 \
  --normalization_method standard \
  --save_plots path_to_output_PDF_file
```

An example output figure is shown in (Figure 2).

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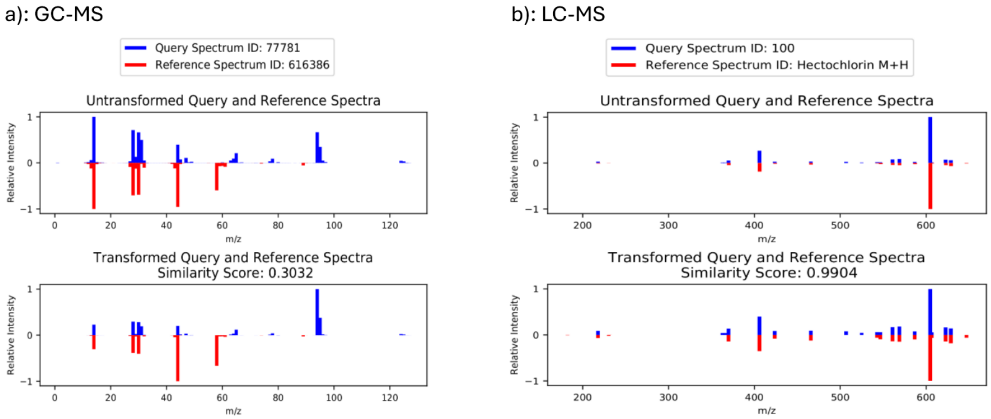

Figure 2: Query spectrum plotted against reference spectrum before/after spectrum preprocessing transformations. A low-entropy transformation is applied to the LC-MS spectra, and a weight factor tranformation is applied to the GC-MS spectra.

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